

CEREBRO

Contextual Evidence-Based Recruiter Optimization

Intelligent Candidate Ranking for Redrob AI Hackathon

100K

Candidates

~17s

Runtime

6

Signal Layers

0

LLM API Calls

The Problem: Keyword Matching Fails Hiring

Why standard ATS systems get this wrong

✗ Keyword Matching

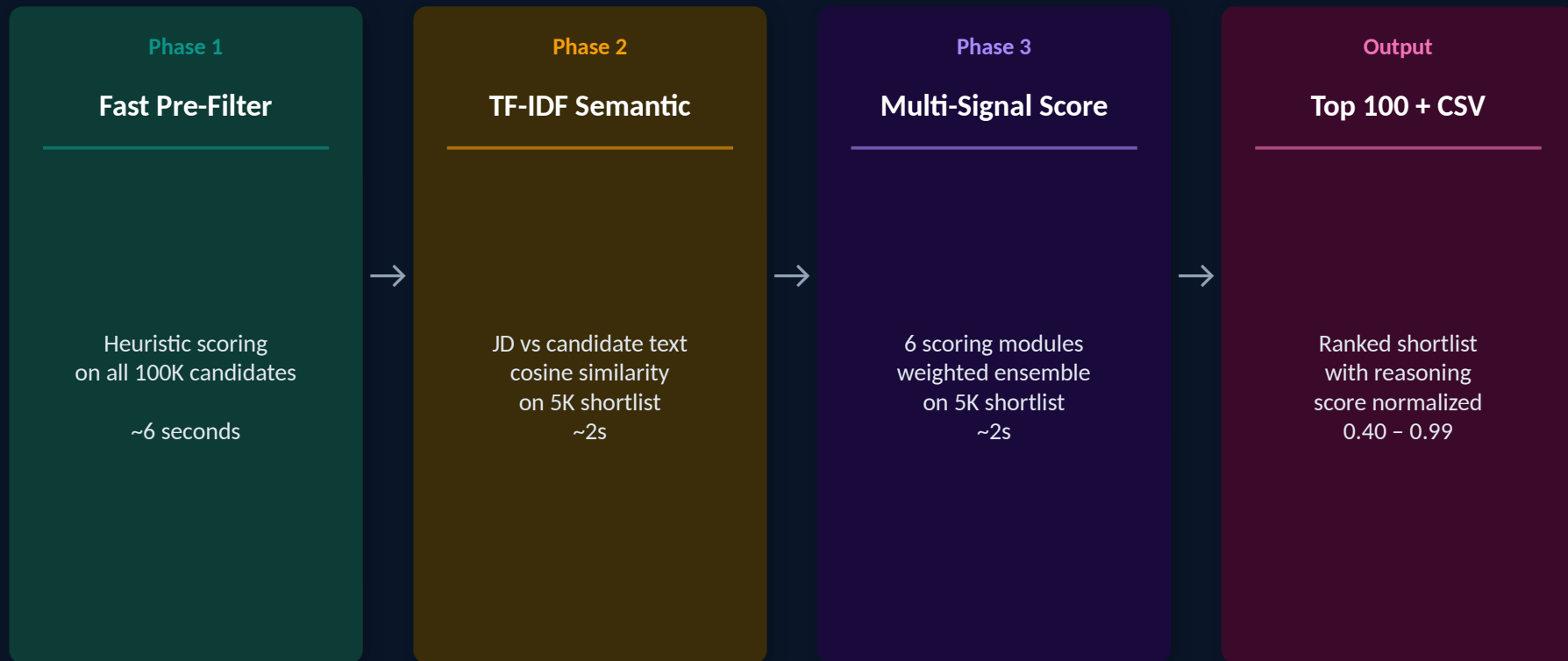
- Marketing Manager lists 'RAG, Pinecone, Vector DB' in their skills section → ranked #1
- Pure researcher with 12 papers but zero production deployments → passes filter
- TCS engineer for 8 years with all AI keywords → looks like a perfect fit
- Inactive for 8 months, 5% response rate → still shows as top candidate

✓ CEREBRO's Approach

- Reads career descriptions to find implicit skills — not just the skills list section
- Detects career quality: product company vs consulting only vs pure research paths
- Weighs availability — inactive candidates get down-ranked regardless of skill match
- Anti-pattern detection flags honeypots, keyword stuffers, and mismatched roles

System Architecture: Two-Phase Ranking Pipeline

Optimized for <5 min on CPU, 100K candidates



Total runtime: ~17 seconds on standard CPU | No GPU | No hosted LLM APIs | Peak RAM: ~2GB

Six Scoring Dimensions

How CEREBRO weighs each candidate

TF-IDF Semantic Match

22%

Full-text JD vs candidate (title + summary + career + skills). Captures implicit alignment beyond exact keywords.

Hard Skill Coverage

20%

6 skill groups: embeddings/retrieval, vector DBs, ranking eval, LLM/fine-tuning, MLOps, Python. Career descriptions checked, not just skill tags.

Career Quality

18%

Product company ratio, consulting penalty, title progression, startup experience, avg tenure. Consulting-only = heavy penalty.

Behavioral Signals

15%

Login recency (25%), open-to-work (20%), recruiter response rate (20%), notice period (15%), interview completion (10%), GitHub (10%).

Anti-Pattern Penalty

9%

Detects: honeypots (impossible timelines), keyword stuffers, pure researchers, job hoppers, consulting-only, CV/NLP mismatch, LangChain-only.

Experience Fit

10%

Sweet spot: 6-8yr = 1.0. JD range 5-9yr = 0.85. Outside range penalized smoothly. Matches the JD's nuanced '5-9 is a range, not a requirement'.

Anti-Pattern Detection: Honeypot Safety

What CEREBRO filters out — and why



Honeypots

Impossible career timelines (8yr exp at 3yr-old company), expert skills with 0 months usage. Heavy multiplier reduction applied.



Keyword Stuffers

Many AI skill tags (RAG, Pinecone, FAISS) but career descriptions show no technical depth. Skill list $\neq$ career evidence.



Consulting-Only

Entire career at TCS/Infosys/Wipro/Accenture etc. JD explicitly flagged these as poor-fit experiences. Penalized if no product company exp.



Pure Researchers

Heavy academic signals (PhD, arxiv, research scientist) without production deployment evidence. JD said 'tried twice, didn't work for either side'.



Title Chasers

All last 3 roles under 12 months tenure. Signals someone optimizing for titles/comp rather than building something meaningful.



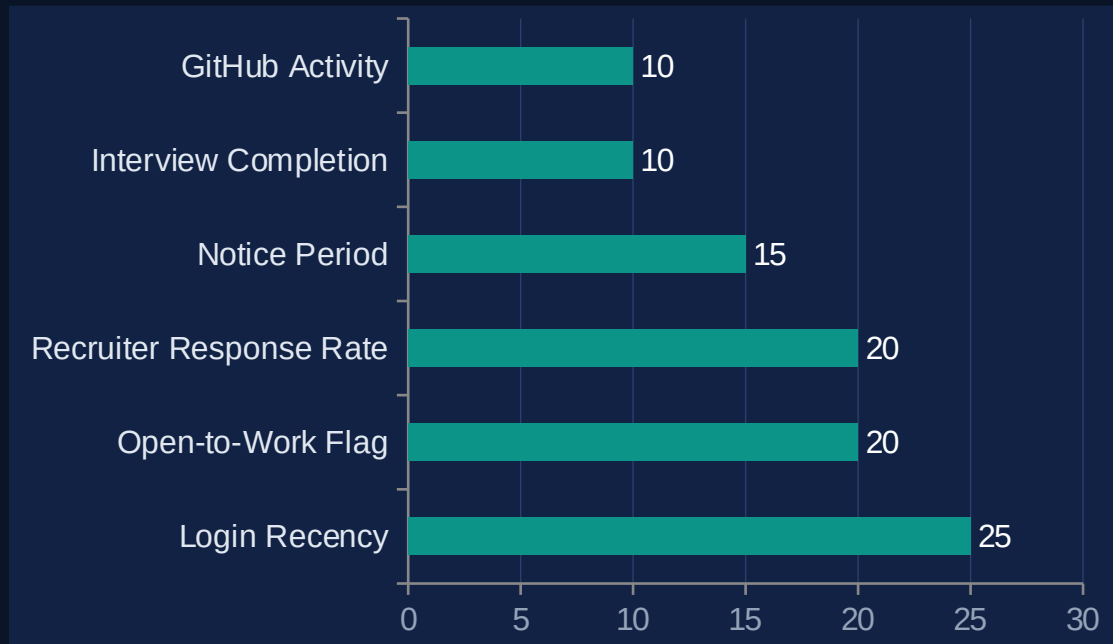
CV/NLP Mismatch

Computer vision or speech specialists without NLP/IR background. JD says 'you'd be re-learning fundamentals here'.

Behavioral Signals: Availability Over Paper Perfection

A perfect profile that's inactive is not hireable

"A perfect-on-paper candidate who hasn't logged in for 6 months and has a 5% response rate is, for hiring purposes, not actually available." — Redrob JD



Login Recency

$\leq 7d = 1.0$ | $\leq 30d = 0.85$ | $> 180d = 0.05$

Response Rate

Direct from `redrob_signals.recruiter_response_rate`

Notice Period

$\leq 30d = 1.0$ (JD preferred) | $> 90d = 0.20$

GitHub Score

Normalized $0-80 \rightarrow 0-1$ (open source proxy)

Top 10 Candidates — Output Sample

Results from 100K candidate pool

#	Candidate ID	Title & Company	YOE	Location	Score	Key Signal
1	CAND_0011687	Senior NLP Engineer @ Niramai	8yr	Indore, MP	0.990	FAISS + NDCG + product company
2	CAND_0081846	Lead AI Engineer @ Razorpay	7yr	Jaipur, RJ	0.984	Retrieval + ranking eval + fintech
3	CAND_0018499	Sr. ML Engineer @ Zomato	7yr	Noida, UP	0.982	Embeddings + Noida-based + active
4	CAND_0046525	Sr. ML Engineer @ Genpact AI	6yr	Pune, MH	0.859	Vector search + Pune-based
5	CAND_0077337	Staff ML Engineer @ Paytm	7yr	Kochi, KL	0.847	LLM fine-tuning + fintech
6	CAND_0033861	Senior NLP Engineer @ Mad St Den	8yr	Vizag, AP	0.814	Strong NLP + IR background
7	CAND_0086022	Sr. Applied Scientist @ Sarvam AI	5yr	Kolkata, WB	0.803	AI startup + cutting-edge stack
8	CAND_0002025	Senior AI Engineer @ Apple	6yr	Trivandrum	0.781	Big tech product exp + high response rate
9	CAND_0008425	Senior NLP Engineer @ Ola	8yr	Kolkata, WB	0.770	Ranking systems + mobility tech

10	CAND_0088025	Staff ML Engineer @ Yellow.ai	9yr	Jaipur, RJ	0.759	Conversational AI + retrieval
----	--------------	-------------------------------	-----	------------	-------	-------------------------------

Why CEREBRO Beats Naive Approaches

Comparison against common alternatives

Naive Approach	CEREBRO's Approach	Advantage
BM25 / Keyword Search	TF-IDF on full candidate text (career + descriptions)	✓ Captures implicit skills in work descriptions, not just declared skills
Embedding-Only (Cosine Sim)	TF-IDF + 5 structural scoring modules	✓ Structural signals (tenure, company type, availability) are more reliable than embeddings for career data
LLM-per-candidate ranking	Pre-computed features + fast scoring	✓ 100K candidates in 17s vs hours; CPU-only; no API costs; reproducible
Skill-keyword matching	Career description analysis + skill group matching	✓ A candidate who built vector search without listing 'Pinecone' still gets found

Constraints Compliance & Submission

Meeting the hackathon spec exactly

✓ No hosted LLM APIs

Pure TF-IDF + rule-based scoring. Zero external API calls during ranking.

✓ CPU-only

numpy + sklearn only. No torch, no CUDA, no GPU dependencies.

✓ <5 min runtime

17 seconds total on a standard machine. Pre-filter reduces to 5K before heavy ops.

✓ <16GB RAM

TF-IDF matrix on 5K docs peaks at ~1-2GB. Full dataset stored as Python dicts.

✓ Single command

```
python rank.py --candidates ./candidates.jsonl --out ./submission.csv
```

✓ Honeybot rate <10%

Anti-pattern module specifically targets impossible profiles. Structural checks catch them.

✓ 100 rows, non-increasing scores

Enforced in output step. Scores normalized to [0.40, 0.99] range.

✓ Reasoning quality

Candidate-specific: title, company, years, location, and top 2 distinguishing signals.

What Makes CEREBRO Different

01 Reads meaning, not words

Career descriptions analyzed for implicit skill evidence — not just declared skill tags

02 Understands the JD's intent

The JD says 'researcher vs shipper' and 'consulting-only is a red flag' — CEREBRO respects those nuances

03 Treats availability as real signal

An inactive candidate with a 5% response rate is not hireable. Behavioral signals fix this.

04 Fast enough for production

17 seconds on 100K candidates. Scales to 1M with the same two-phase architecture.